

ECEN-662: Homework 2

Due on February 6, 2026 at 11:59 PM

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Problem 1

i) Let

$$f_{\theta}(x) = h(x) \exp(\eta^t T - A(\eta))$$

be an s -parameter exponential family of distributions. Show that if there exists

$$\{\theta_0, \theta_1, \dots, \theta_s\} \subseteq \Theta$$

such that the vectors

$$(\eta(\theta_i) - \eta(\theta_0) : i \in [s])$$

are linearly independent in $\mathcal{R}^s$, then T is a minimal sufficient statistic.

ii) Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is the unknown parameter. Use the result of part i) to show that

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is a minimal sufficient statistic.

Solution

Problem 2

Suppose that T_1 is sufficient and T_2 is minimally sufficient. Let U be an unbiased estimator of θ and define

$$U_1 = \mathbb{E}_\theta[U|T_1] \quad \text{and} \quad U_2 = \mathbb{E}_\theta[U|T_2]$$

Show that $U_2 = \mathbb{E}_\theta[U_1|T_2]$ and hence $\text{Var}_\theta[U_2] \leq \text{Var}_\theta[U_1]$.

Solution

Problem 3

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{U}([0, \theta])$, and $\theta > 0$ is the unknown parameter. Suppose g is a differentiable function on $\mathcal{R}_{++}$. Find an UMVUE of $g(\theta)$ in terms of the function g .

Solution

Problem 4

Let $X = [X_1, \dots, X_n]^t$, where

$$X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda),$$

and $\lambda > 0$ is the unknown parameter. Suppose we wish to estimate

$$\theta = \exp(-\lambda) = \mathbb{P}_\lambda(X_i = 0).$$

- i) Show that $T = \sum_{i=1}^n X_i$ is a complete sufficient statistic for this likelihood family.
- ii) Show that $S_1 = 1_{\{X_1=0\}}$ and $S_2 = \frac{1}{n} \sum_{i=1}^n 1_{\{X_i=0\}}$ are both unbiased estimate of θ .
- iii) Compute the Rao-Blackwellized estimators $\tilde{S}_i = \mathbb{E}_\theta[S_i|T]$, $i = 1, 2$. What does your answer have to do with the completeness of T ?

Solution